MIL-P-54001A

9 March 1973
SUPERSEDING
MIL-P-54001
25 April 1962

MILITARY SPECIFICATION

PLUGS, PIPE, MAGNETIC, HEADLESS: IRON, STEEL OR ALUMINUM ALLOY

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers: (1) headless magnetic pipe plugs having recessed hex heads, up to and including one inch size, conforming to SAE Standard J531, and, (2) having recessed square heads, for sizes over one inch, conforming to ANSI Standard B16.14. All plugs are specified on MS49006.

1.2 Classification. Magnetic pipe plugs shall be of the following types:

Type I. Aluminum Alloy

Type II. Malleable Iron, Cast Iron, or Cold Finished Steel.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

FSC 4730

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SPECIFICATIONS

Federal

QQ-A-225/1	Aluminum Alloy Bar, Rod, and Wire; Rolled, Drawn, or Cold Finished, 1100.
QQ-A-225/3	Aluminum Alloy Bar, Rod, and Wire; Rolled, Drawn, or Cold Finished, 2011.
QQ-A-225/6	Aluminum Alloy Bar, Rod and Wire; Rolled, Drawn, or Cold Finished, 2024.
QQ-A-225/8	Aluminum Alloy Bar, Rod Wire, and Special Shapes; Rolled, Drawn, or Cold Finished, 6061.
QQ-A-367	Aluminum Alloy Forgings.
QQ-I-652	Iron Castings, Grey.
QQ-I-666	Iron, Malleable, Ferritic, for Castings.
QQ-S-634	Steel, Bar, Carbon, Cold Finished (Standard Quality).
QQ-S-681	Steel Castings.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Boxes, Shipping, Fiberboard.
PPP-B-640	Boxes, Fiberboard, Corrugated, Triplewall.
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper (For Carton Sealing).

Military

MIL-P-116	Preservation, Methods of
MIL-P-3420	Packaging Materials, Volatile Corrosion Inhibitor Treated, Opaque.
MIL-A-8625	Anodic Coatings, For Aluminum and Aluminum Alloys.

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Standards

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipping and Storage
MS49006	Plugs, Pipe, Magnetic, Headless, Aluminum, Iron or Steel

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Bureau of Standards

Handbook H28 Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402).

Standard Specification for Permanent Magnet Materials

MMPA-0100 Permament Magnet Material

(Application for copies should be addressed to the Magnetic Material Producers Association, 3525 W. Peterson Avenue, Chicago, IL 60645.)

American Trucking Association, Inc.

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, Inc. 1616 P. Street, NW., Washington, DC 20036.)

Uniform Classification Committee

Uniform Freight Classification

(Application for copies of these ratings, rules and regulations should be addressed to Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, IL 60606.)

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Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Material.

3.1.1 Malleable iron. Material for malleable iron plugs shall be in accordance with QQ-I-666, grade 1 or 2.

3.1.2 Cast iron. Material for cast iron plugs shall be in accordance with QQ-I-652, class 20 through class 30.

3.1.3 Cold finish steel. Material for cold finish steel plugs shall be in accordance with QQ-S-634 or QQ-S-681.

3.1.4 Aluminum. Material for aluminum alloy plugs shall be in accordance with QQ-A-225/1, QQ-A-225/3, QQ-A-225/6, QQ-A-225/8 or QQ-A-367.

3.2 Fabrication. Plugs may be fabricated by casting, forging, or machining.

3.3 Design. The magnetic pipe plug design requirements for dimensions and tolerances, magnetic lift, and pipe thread data shall be in accordance with MS49006.

3.4 Finish.

3.4.1 Type I. Aluminum alloy plugs shall be anodized in accordance with MIL-A-8625, type I, class 1, or type II, class 1.

3.4.2 Type II. Ferrous plugs shall be plain (uncoated) finish.

3.5 Magnet. The magnet shall be a permanent magnet in accordance with Magnetic Materials Producers Association Standard MMPA-0100.

3.5.1 Attachment. The magnet shall be attached in such manner as to require physical displacement of metal before the magnet can be removed. Brazing, welding, staking, or swedging are acceptable methods of attaching.

3.6 Workmanship. Plugs shall be free from burrs, cracks, corrosion, deformations, or other defects causing the item to be unsuitable for the purpose intended.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Lot. A lot shall consist of all plugs of the same type and size offered for delivery at one time.

4.3 Sampling.

4.3.1 Sampling for examination. Sampling for examination shall be in accordance with MIL-STD-105.

4.3.2 Sampling for tests. Sampling for tests shall be in accordance with MIL-STD-105, Inspection Level S1.

4.4 Examination. Each sample plug selected in accordance with 4.3.1 shall be visually examined for the following defects to insure product conformance. Lot acceptance or rejection shall be based on an AQL of 2.5 percent defective for major defects and an AQL of 6.0 percent defective for minor defects.

4.4.1 List of defects.

Categories	Defects
Critical	None defined
Major	
101	Cold shuts
102	Blow holes
103	Porosity
104	Adherent sand or scale
105	Dimensions, tolerances, or pipe thread not in accordance with MS49006
106	Magnet strength less than specified
107	Loose magnets
108	Improper magnet attachment (see 3.5.1)
Minor	
201	Rough finish
202	Rough core

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4.4.2 Dimensions. Plugs in which limited small areas are slightly under the specified minimum metal thickness may be considered acceptable under this specification, provided such areas are at no point less than 90 percent of the specified thickness, and provided the fittings fully meet all other requirements.

4.5 Tests. Each sample plug selected in accordance with 4.3.2 shall be tested as specified in 4.5.1, followed by testing as specified in 4.5.2 if plugs are of the malleable iron type. Lot acceptance or rejection shall be based on a AQL of 4.0 percent defective for each test.

4.5.1 Hydrostatic pressure test. Each plug shall be subjected to a hydrostatic pressure test of 250 pounds per square inch for one minute. The plug shall not leak water. Plugs used for this test shall not be included in the shipment to the Government.

4.5.2 Soundness test. Malleable iron plugs shall be hammered or crushed to determine whether the heat treatment has been properly performed. Failure to bend so that plastic deformation is readily apparent to the naked eye, or a showing of a typical malleable fracture, shall be considered evidence of unsatisfactory annealing.

4.6 Rejected lots. Disposition of rejected lots shall be in accordance with MIL-STD-105.

4.7 Defective units. Units found defective in the course of inspection shall be corrected to the satisfaction of the Government or shall be replaced with non defective units.

4.8 Preservation, packaging and packing. Preservation, packaging and packing of the plugs shall be examined to determine compliance with the requirements of Section 5 of this specification, and as required by specifications referenced therein. Except as required by referenced specifications, inspection lots and sampling shall be in accordance with MIL-STD-105, at inspection level S4 and an AQL of 4.0 percent defective.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be for Level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Plugs shall be cleaned Process C-1 of MIL-P-116.

5.1.1.2 Drying. Plugs shall be dried by any applicable process of MIL-P-116 non-injurious to the item.

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le drop selected in accordance with 4.4 Examination. Each samp

5.1.1.3 Preservation. Unless otherwise specified in the contract or order, application of a preservative shall be in accordance with 5.1.1.3.1 and 5.1.1.3.2.

5.1.1.3.1 Type I. No preservation required.

5.1.1.3.2 Type II. Preservation P-18 of MIL-P-116 required.

5.1.1.4 Unit packaging.

5.1.1.4.1 Quantity. Unless otherwise specified in the contract or order (see 6.2), identical plugs shall be packaged in numerical quantities specified herein. Nominal sizes one inch and under shall be unit packaged in quantities of fifty each. Nominal sizes over one inch shall be unit packaged in quantities of ten each.

5.1.1.4.2 Unit packaging Type I. Type I plugs shall be unit packaged Method III of MIL-P-116. The unit container shall be a minimum size PPP-B-636 fiberboard box, class weather resistant. Closure shall be accomplished by sealing all seams and joints with PPP-T-76 tape, two inches wide.

5.1.1.4.3 Unit packaging Type II. Type II plugs shall be unit packaged Method I of MIL-P-116. The unit container shall be a minimum size PPP-B-636 fiberboard box, class weather resistant. The container shall be lined with VCI treated material conforming to Type I, Class 3, Style A of MIL-P-3420. Closure shall be accomplished by sealing all seams and joints with PPP-T-76 tape, two inches wide.

5.1.2 Level C Unit packaging.

5.1.2.1 Level C. Plugs shall be preserved and packaged to afford adequate protection against corrosion, deterioration, and damage during shipment from the supply source to the receiving activity. The supplier may use his standard practice when it meets these requirements.

5.1.3 Intermediate packaging. Not applicable.

5.2 Packing. Packing shall be Level A, B, or C as specified (see 6.2).

5.2.1 Level A. Unit packages, as specified in 5.1.1.4, shall be packed in a minimum size exterior box conforming to Stype A, B, I or J of PPP-B-601, Class 2 of PPP-B-621, or Class 3 of PPP-B-585. Unless otherwise specified, the gross weight of the container shall not exceed 200 pounds. Closure of the container shall be in accordance with the applicable container specification.

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5.2.2 Level B. Unit packages, as specified in 5.1.1.4, shall be packed in a minimum size exterior container conforming to Class 2 of PPP-B-640, class weather resistant of PPP-B-636, Class 1 of PPP-B-631, type domestic of PPP-B-601, or Class 1 or 2 of PPP-B-585. Unless otherwise specified, the gross weight of the container shall not exceed the limitations of the container specification. Closure shall be accomplished in accordance with the requirements of the applicable container specification.

5.2.3 Level C. The plugs shall be packed in a manner which shall insure arrival at destination in satisfactory condition and be acceptable to the carrier at lowest rates. Containers and packing shall comply with Uniform Freight Classification Rules and National Motor Freight Classification Rules.

5.3 Marking. Interior and exterior containers shall be marking in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Intended use. Magnetic plugs are intended for use in automotive engine oil systems, transmissions, gear boxes, circulating oil systems and other systems of circulating fluids in which ferrous metal particles or chips may be present.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) MS part number of plug (see 3.3).

(c) Level of packaging and packing required (see 5.1 and 5.2).

CUSTODIANS:
Army - WC
Navy - YD
Air Force - 82

REVIEW ACTIVITIES:
Army - ME
Navy - SH
Air Force -
DSA - CS
USER ACTIVITIES:
Army - AT - AV - EL - GL
Navy - MC
Air Force -

PREPARING ACTIVITY:
Army - WC

Project No. 4730-0069

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

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